

Hybrid Routing Agent

Every query takes the cheapest path that still gets the answer right.

local-first · \$0

fireworks fallback

12-signal router

■ LOCAL

phi3:mini · 0
tokens

▲ score < 0.30 · ▼ score > 0.65

◆ FIREWORKS

only when needed

Accuracy is the gate. Tokens are the score.

STEP 1 — GATE

An LLM judge checks every answer.
Fail the accuracy bar → **off the
leaderboard.**

STEP 2 — RANK

Survivors ranked by total Fireworks
tokens. **Fewest tokens wins.**

💡 Most prompts don't need a frontier model. Paying big-model tokens to answer " $12 + 47$ " is throwing away rank — **route it to a model that costs nothing.**

A 12-signal complexity router

Regex + keyword signals (code, math, reasoning, NER-as-JSON, sentiment, format constraints...) score every prompt 0→1. No extra LLM call — routing itself costs zero tokens.

< 0.30 → **LOCAL** phi3:mini, free

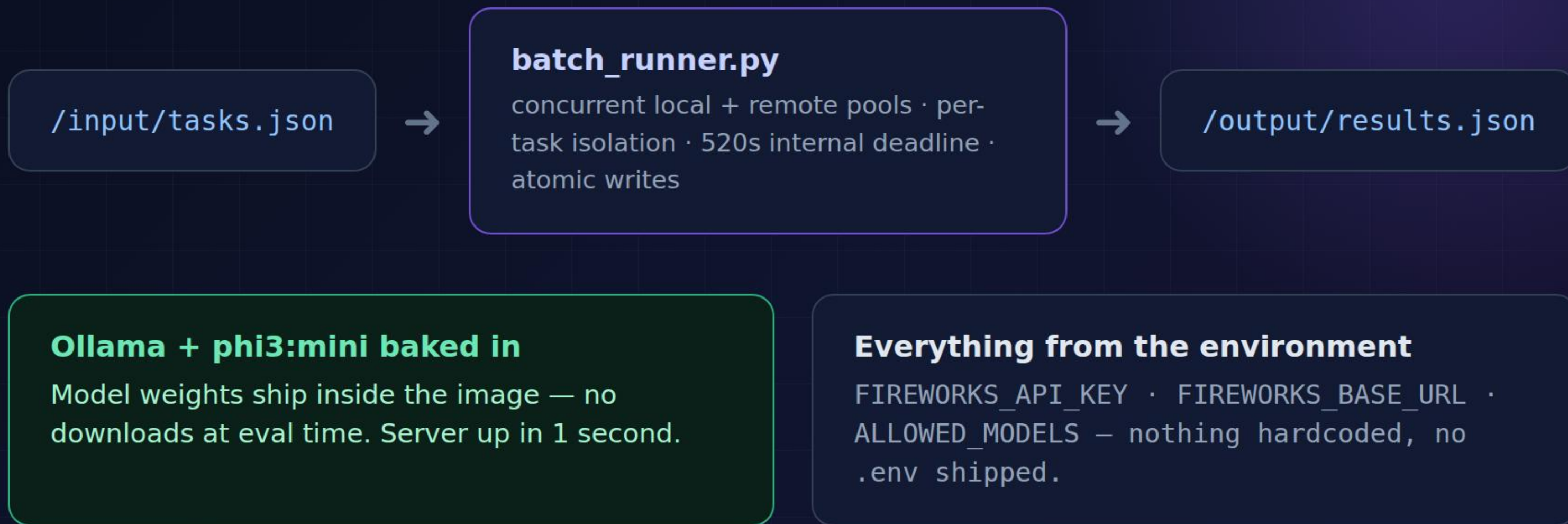
0.30-0.65 → **CASCADE** try local,
escalate

> 0.65 → **REMOTE** Fireworks AI

Cascade: a confidence heuristic inspects the local answer — hedging, length, repetition. Confident → keep it free. Shaky → escalate.

Remote: fast model by default; the strong model only for complexity ≥ 0.85 . Both picked at runtime from `ALLOWED_MODELS`.

One container, zero setup



Verified in CI on the published image

0

remote tokens on the full
sample suite — 12/12
correct, all local

8 / 8

capability categories
answered end-to-end
inside the container

50s

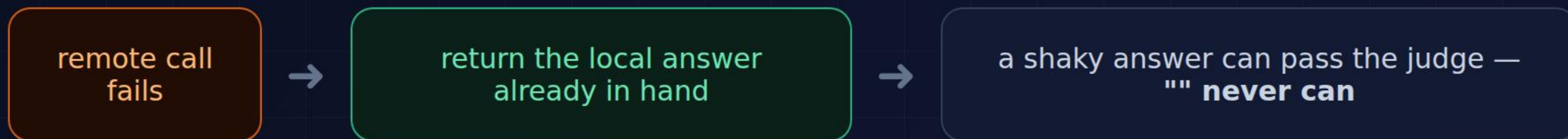
full batch wall-clock —
the limit is 10 minutes

3.6GB

compressed image —
36% of the 10 GB cap

GitHub Actions builds the exact submitted image, checks the size cap, runs it against sample tasks for all 8 categories, and validates the results contract — **on every push.**

An empty answer is a guaranteed miss



- ✓ One task crashing can't sink the batch
- ✓ Deadline hit → partial results, still valid JSON
- ✓ Atomic writes — never a half-written file

- ✓ Ollama down → clean remote-only degradation
- ✓ Fast/strong models chosen from ALLOWED_MODELS at runtime
- ✓ Every failure path exercised in CI

Big-model accuracy. Small-model bill.

```
docker pull ghcr.io/syk-v1/hybrid-routing-agent:latest
```

github.com/syk-v1/hybrid-routing-agent · Ollama phi3:mini + Fireworks AI · AMD Developer Hackathon Track 1